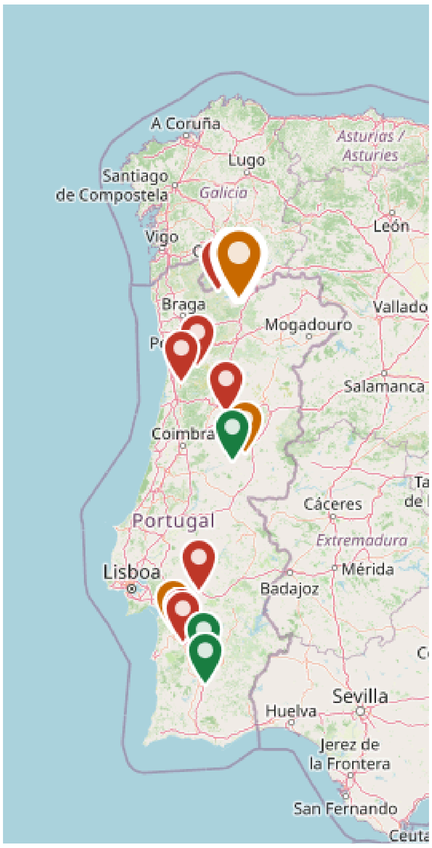
Water Impacts due to Mining in Portugal

Data analysis of water pollution and depletion related to mining 2015-2016



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List of acronyms

The following acronyms are used throughout the report, listed in alphabetical order.

Acronym	Full form and meaning
ACT	Autoridade para as Condicoes do Trabalho - Portuguese Authority for Working Conditions
AIA	Avaliacao de Impacte Ambiental - Environmental Impact Assessment (procedure/process)
AMD	Acid mine drainage
APA	Agencia Portuguesa do Ambiente - Portuguese Environment Agency
APRH	Associacao Portuguesa dos Recursos Hidricos - Portuguese Water Resources Association
CRM	Critical raw material
CRMA	Critical Raw Materials Act (EU Regulation 2024/1252)
CTP	Chemical Treatment Plant (at Urgeirica mine)
DCAPE	Decisao sobre a Conformidade Ambiental do Projeto de Execucao - Decision on Environmental Conformity of the Execution Project
DGS	Direccao-Geral da Saude - Directorate-General for Health (Portugal)
DIA	Declaracao de Impacte Ambiental - Environmental Impact Declaration (final regulatory decision issued by APA)
DL	Decreto-Lei - Decree-Law (Portuguese legislative instrument)
DR	Diario da Republica - Portuguese Official Gazette
EDP	Energias de Portugal - Portuguese energy utility company
EDM	Empresa de Desenvolvimento Mineiro - Mining Development Company (state-owned entity responsible for remediation of abandoned mines in Portugal)
EIA	Environmental Impact Assessment (study/report); in Portuguese: Estudo de Impacte Ambiental
ENU	Empresa Nacional de Uranio - National Uranium Company (former Portuguese state entity, now dissolved)
EU	European Union
FAO	Food and Agriculture Organization of the United Nations
FoEE	Friends of the Earth Europe
GDELT	Global Database of Events, Language and Tone - global news monitoring platform used in data collection
GEOTA	Grupo de Estudos de Ordenamento do Territorio e Ambiente - Portuguese environmental NGO
GPSA	Grupo de Pessoas pela Salvaguarda da Argemela - civic group opposing the Argemela mine project

Acronym	Full form and meaning
IGM	Instituto Geologico e Mineiro - Geological and Mining Institute (Portugal, predecessor of LNEG)
LNEG	Laboratorio Nacional de Energia e Geologia - National Laboratory for Energy and Geology (Portugal)
MP	Ministerio Publico - Public Prosecutor / Public Prosecution Service (Portugal)
MW	Megawatt
NGO	Non-governmental organisation
PDAC	Prospectors and Developers Association of Canada - major international mining conference held annually in Toronto
PDM	Plano Diretor Municipal - Municipal Master Plan (Portuguese land-use planning instrument)
PIN	Projeto de Interesse Nacional - Project of National Interest (Portuguese investment classification)
PSD	Partido Social Democrata - Social Democratic Party (Portugal)
RECAPE	Relatorio de Conformidade Ambiental do Projeto de Execucao - Environmental Conformity Report of the Execution Project
REMPC	Regulamento Europeu das Materias-Primas Criticas - see CRMA
RH6	Regiao Hidrografica 6 - Hydrographic Region 6 (Sado and Mira river basins, Portugal)
RTP	Radio e Televisao de Portugal - Portuguese public broadcaster
SALMI	Companhia de Sal Mineiro - Portuguese rock salt mining company operating the Loule mine
SERP	Search engine results page
SIC	Sociedade Independente de Comunicacao - Portuguese private television broadcaster
SIPAM	Sistema Importante do Patrimonio Agricola Mundial - Globally Important Agricultural Heritage System; UN-FAO designation held by the Barroso region
TJUE	Tribunal de Justica da Uniao Europeia - Court of Justice of the European Union (CJEU)
U3O8	Triuranium octoxide - the most stable form of uranium oxide, the primary product of uranium mining
UDCB	Associacao Unidos em Defesa de Covas do Barroso - civic association opposing the Barroso lithium mine
UN	United Nations
WFD	Water Framework Directive (EU Directive 2000/60/EC) - establishes a framework for EU water policy and quality standards
ZERO	Associacao Sistema Terrestre Sustentavel - Portuguese environmental NGO

Glossary

The following terms are used in the context of mining environmental impacts and Portuguese regulatory frameworks, listed in alphabetical order.

Term	Definition
Abandoned mine / Legacy mine	A mine that has ceased operations, often before modern environmental standards were enacted, leaving unresolved liabilities including tailings, acid drainage and contaminated soils. In Portugal, EDM is the state entity responsible for the environmental management and remediation of abandoned mines.
Acid mine drainage (AMD)	Acidic water produced when sulphide minerals in mine waste or exposed rock are oxidised in the presence of water and oxygen. The resulting leachate carries heavy metals (arsenic, lead, zinc, copper) and has a characteristically low pH (sometimes below 2). AMD is the primary water quality concern at pyrite belt mines such as Neves-Corvo, Aljustrel, Lousal, Caveira and Sao Domingos.
Aquifer	An underground layer of permeable rock or sediment that stores and transmits groundwater. Several mine projects in Portugal - notably Barroso, Gandarela, Lagoa Salgada and Argemela - raised concerns about the impact of extraction on local aquifers used for domestic and agricultural water supply.
Concessao mineira (Mining concession)	A legal right granted by the Portuguese state authorising the exploration and/or extraction of mineral resources in a defined area. Mining concessions in Portugal are governed by the Lei de Bases do Mineracao (Law 54/2015) and administered by the DGEG (Direcao-Geral de Energia e Geologia).
Critical raw material (CRM)	A raw material identified by the European Commission as economically important to the EU economy and subject to supply risk. The current EU list (2023) includes 34 materials. Several mines in this report extract CRMs: Neves-Corvo and Aljustrel (copper), Panasqueira and Borralha (tungsten), and Barroso, Romano de Tresminas, Argemela and Gandarela (lithium).
Declaracao de Impacte Ambiental (DIA)	The final decision issued by the APA at the conclusion of an environmental impact assessment procedure in Portugal. A conditional DIA imposes specific obligations on the developer (e.g. protection of watercourses, monitoring requirements). A negative DIA, such as that issued for Lagoa Salgada in January 2026, rejects the project.
Escombreira (Waste rock dump)	A surface deposit of waste rock excavated during mining operations but not processed for ore. Escombreiras can generate acid drainage and dust, and pose risks of slope instability. They are a persistent visual and environmental legacy at abandoned mines such as Panasqueira, Urgeirica and Sao Domingos.
Estudo de Impacte Ambiental (EIA)	The environmental impact assessment study submitted by a project developer in Portugal, setting out the anticipated environmental impacts and proposed mitigation measures. Distinct from the DIA, which is the regulatory decision issued by APA on that study.
Faixa Piritosa Iberica (FPI) / Iberian Pyrite Belt	A geological formation of approximately 250 x 50 km spanning the Alentejo region of Portugal and the Huelva region of Spain, recognised as

Term	Definition
	the world largest province of volcanogenic massive sulphide (VMS) deposits. It hosts the Neves-Corvo, Aljustrel, Lousal, Caveira, Sao Domingos and Lagoa Salgada mines documented in this report.
Heavy metals	Metallic elements with relatively high density that are toxic at low concentrations. Those most commonly associated with mining in Portugal include arsenic (As), lead (Pb), zinc (Zn), copper (Cu), cadmium (Cd) and mercury (Hg). Elevated concentrations of these elements in water constitute a breach of the EU Water Framework Directive and Portuguese water quality standards.
Lepidolite	A lithium-rich mica mineral that is one of the primary sources of lithium in the Romano de Tresminas project in Montalegre. Lepidolite processing generates significant quantities of tailings with potential environmental impacts.
Nivel freatico (Water table)	The upper surface of the zone of saturation in an aquifer. Open-pit mining operations can lower the water table through dewatering, affecting nearby wells, springs and watercourses. Concern about water table depletion is central to the opposition to the Barroso and Gandarela lithium projects.
Patrimonio Agricola Mundial (SIPAM)	Globally Important Agricultural Heritage System - a designation by the UN-FAO recognising agricultural landscapes of outstanding importance. The Barroso region holds this designation, which is at the core of legal challenges to the Barroso lithium mine, with courts and the public prosecutor arguing that open-pit mining is incompatible with SIPAM status.
pH	A logarithmic scale (0 to 14) measuring the acidity or alkalinity of water. A pH of 7 is neutral; values below 7 are acidic. The EU Water Framework Directive requires surface water pH to remain between 6 and 9 for good ecological status. Acid mine drainage from the Sao Domingos mine has been recorded at pH as low as 2.0.
Rejeitos / Tailings	The finely ground waste material left after valuable minerals have been extracted from ore. Tailings are typically deposited in tailings dams or impoundments. They can contain residual heavy metals, processing chemicals (including cyanide, used in gold processing) and acid-generating sulphides, posing long-term water contamination risks.
Remediacao ambiental (Environmental remediation)	Actions taken to reduce or eliminate contamination at a polluted site, restoring it to a safe condition. In Portugal, EDM is responsible for the remediation of abandoned mining areas under state mandate. Works include capping of tailings, drainage systems, mine water treatment and revegetation.
Spodumene	A lithium aluminium silicate mineral that is the principal lithium ore targeted by the Barroso Lithium Project in Boticas. The Barroso deposit is considered the largest known spodumene deposit in Europe.
Tailings dam (Barragem de rejeitos)	An engineered structure used to contain mining waste (tailings) - the residual slurry left after ore processing. Failures or partial collapses of tailings dams can release toxic material into watercourses. The partial collapse of an inactive tailings dam at Panasqueira in January 2026 is the most recent such incident documented in this report.
Volfâmio / Tungsten (W)	A hard, dense metallic element classified as a critical raw material by the EU. Portugal is one of Europe major tungsten producers, with Panasqueira (operated by Beralt Tin and Wolfram / Almonty Industries) as the primary active mine. Borralha was also a historic tungsten producer.

Term	Definition
Water Framework Directive (WFD)	EU Directive 2000/60/EC establishing a framework for the protection of inland surface waters, transitional waters, coastal waters and groundwater. It requires member states to achieve good ecological and chemical status for all water bodies. Several incidents and projects in this report involve alleged non-compliance with WFD standards, particularly regarding heavy metal concentrations and water quantity.

Executive summary

This report documents water-related infringements caused by mining activities in Portugal - (alleged) non-compliance with regulations/standards due to water contamination and/or water depletion. The monitoring period covers 2015 to 2026. Mines are grouped into three categories: *1) Active mines - currently in operation, 2) Planned - projects under development (awaiting licenses), 3) Abandoned - closed mines with legacy environmental impacts.*

For each mine, the report presents: (1) the name and operator of the mining project; (2) the location; (3) the type of critical raw material (CRM) or mineral extracted; (4) the date(s) of documented incidents; (5) a description of water contamination or depletion issues, including references to breaches of water quality standards; and (6) the sources consulted.

The report is based on automated monitoring of Portuguese media, official documents, academic publications, NGO communications and legal records. Monitoring was conducted using a pipeline that searched for mining-related environmental incidents across multiple data sources (DataForSEO SERP, GDELT DOC 2.0, and Portuguese media RSS feeds), followed by automated classification using keyword patterns aligned with EU environmental law.

The five most extensively documented cases are: Barroso (162 articles), Romano de Tresminas (83), Neves-Corvo (82), Aljustrel (54) and Lagoa Salgada (20). The most recent significant incident recorded was the collapse of a tailings dam at Panasqueira in January 2026, with toxic waste reaching the Zezere river, a key water supply source for Lisbon. The Lagoa Salgada project was rejected by the APA in January 2026 on explicit water resource grounds. The Barroso lithium project is currently before the EU Court of Justice (TJUE), challenged by local associations and ClientEarth on environmental grounds.

The data documented in this report suggest that Portugal's mining sector, both legacy operations and new mining projects, presents a persistent and growing challenge to compliance with EU and national water protection and environmental standards. Mining projects may particularly lead to infringements with the Water Framework Directive, while compliance with the EIA Directive often shows irregularities. The available evidence, from the Panasqueira tailings dam collapse, to the Barroso legal challenge, to the still unremediated mining legacy of Azeméis, suggests that in practice, economic and industrial policy objectives have frequently prevailed over environmental protection required by EU and national environmental law.

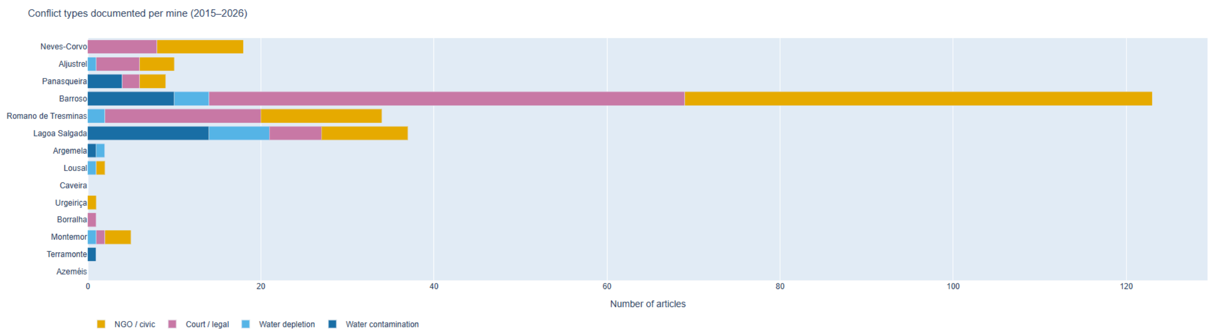
As a result, the cumulative cost of remediating Portugal's 199 abandoned mines has exceeded 200 million euros over 25 years of EDM intervention, with ongoing works financed through successive EU structural fund programmes (QCA III, FEDER, Norte 2030, Portugal 2030).
The volume of civic and legal activity documented in this report, reflects the degree to which affected communities and environmental organisations have been forced to resort to litigation and mobilisation due to inadequate regulatory oversight.

General analysis

Portugal's mining sector occupies an increasingly contested position at the intersection of two competing imperatives of European policy: the drive to secure domestic supplies of raw materials (now accelerated by the EU Critical Raw Materials Act) of 2024, and the obligation to protect water bodies under the EU Water Framework Directive (WFD, Directive 2000/60/EC) and to subject extractive projects to rigorous environmental impact assessment under the EIA Directive (2011/92/EU). The 14 mine sites documented in this report, spanning active operations, projects under licensing and abandoned legacy sites, together illustrate how these imperatives are colliding in practice, and how existing legal frameworks to protect water bodies from mining-related harm are not adequately enforced.

The Water Framework Directive in a mining context

The WFD obliges member states to achieve or maintain good ecological and chemical status in all water bodies. Several of the sites documented here present persistent or potential non-compliance with that objective. On the Iberian Pyrite Belt, the Aljustrel and Panasqueira mines, both active operations, generate acid mine drainage that carries lead, zinc, arsenic and copper into watercourses feeding the Sado and Zêzere basins, respectively. In January 2026, heavy rains caused the partial collapse of a tailings dam at Panasqueira, releasing toxic waste into the Cebola stream, a tributary of the Zêzere river, which supplies the Castelo do Bode reservoir serving Lisbon and dozens of downstream municipalities. Water abstraction for local supply was temporarily suspended. The company had publicly acknowledged, the previous year, that it was discharging untreated effluent into a Zêzere tributary, a practice that is not in line with WFD requirements.



Among abandoned sites, the pattern is more diffuse but no less serious. The Azeméis mine (Minas do Pintor), which was closed in 1995 without any remediation, continues to generate acid drainage from flooded mine shafts carrying heavy metals, including arsenic, into the Ribeira do Pintor and the Antuã river. Terramonte discharged lead, zinc and silver from tailings into the Ribeira de Terramonte and ultimately the Douro, a transnational river system. At both sites, remediation by EDM (Empresa de Desenvolvimento Mineiro) has been initiated only in the 2000s and 2010s, in some cases decades after abandonment. The WFD's requirement for good water status does not exempt historical liabilities; member states are also obliged to address legacy contamination.

Environmental Impact Assessment: approvals under pressure, conditions unenforced

The EIA process, conducted in Portugal by the Agência Portuguesa do Ambiente (APA), is the principal instrument through which mining projects are assessed for their environmental acceptability. The monitoring period documents both the limits of EIA as a protective tool and, in at least one case, its effective use.

FIGURE 1: Environmental relevant events by mine (2015 - 2026). Interactive chart available at: https://sergiofspedro.github.io/Miningeurope/

The Barroso Lithium Project received a conditional EIA approval from APA in May 2023. The conditions included explicit requirements to protect watercourses and groundwater in the Tâmega and Cávado basins. However, the Ministério Público (Public Prosecutor) subsequently requested the annulment of that declaration, arguing it was legally flawed and incompatible with the area's status as a UN-FAO Globally Important Agricultural Heritage System. Three parallel legal proceedings followed, and in February 2026, the association UDCB and the organisation ClientEarth lodged a challenge at the EU Court of Justice. The EIA conditions, including the establishment of a mandatory environmental oversight commission, remained unenforced as of mid-2025. The Romano de Tresminas project followed a similar trajectory: a conditional EIA in 2023, five mining concession contracts signed in late 2024, and an oversight commission that had not been established by the time this report was prepared. In both cases, the existence of an EIA approval with water-protection conditions has not translated into adequate oversight mechanisms, a structural gap that remains problematic .

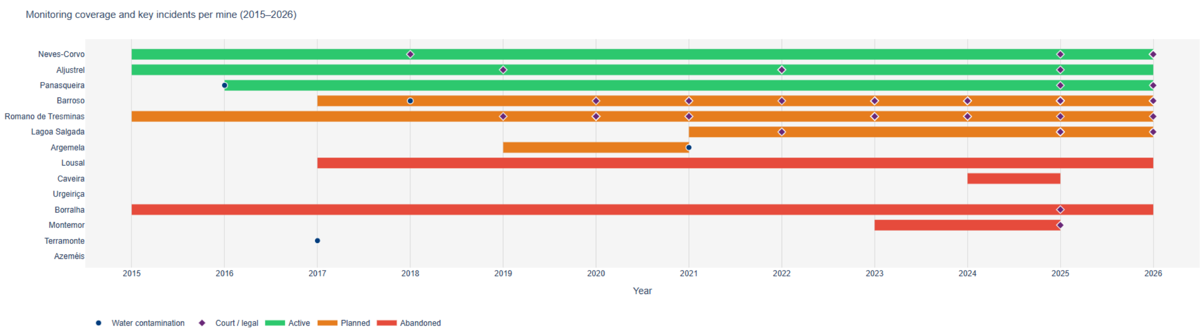


FIGURE 2: Monitoring coverage and key incidents by mine (2015 - 2026). Interactive chart available at: https://sergiofspedro.github.io/Miningeurope/

The Lagoa Salgada case offers a contrasting example. In January 2026, APA issued a negative EIA decision, the first time a major Portuguese mining project was rejected explicitly on grounds of water resource risk. The project would have required 864 cubic metres of groundwater per day in one of Portugal's driest hydrological regions, used sodium cyanide in ore processing, and situated its operations within the same area as the sole water abstraction point serving local communities. The decision was welcomed by environmental groups but contested by the project promoter, which indicated it intended to appeal or resubmit.

The CRMA acceleration and its legal tensions

The EU Critical Raw Materials Act classifies lithium, tungsten, copper and several other minerals as strategic, and requires member states to streamline licensing processes for extraction projects. Three of the four planned projects in this dataset, Barroso, Romano de Tresminas and Argemela, target critical raw materials, as do two of the three active operations (Panasqueira for tungsten, Neves-Corvo and Aljustrel for copper and zinc).Two lithium projects (Barroso, Romano de Tresminas) were classified as strategic under the CRMA.

The CRMA's objective to fast-track mining projects creates a structural tension with EIA and WFD obligations. Accelerated permitting reduces the space for rigorous environmental assessment, public participation and judicial procedures, the very mechanisms through which water protection standards are operationalised. The volume of civic and legal activity documented in this report, reflects the degree to which affected communities and environmental organisations have been forced to compensate through litigation and mobilisation for what they regard as inadequate regulatory scrutiny at the licensing stage. A challenge at the EU Court of Justice against the Barroso project constitutes the most direct legal confrontation between CRMA acceleration and the substantive environmental obligations of EU member states, and its outcome will likely have consequences beyond Portugal.

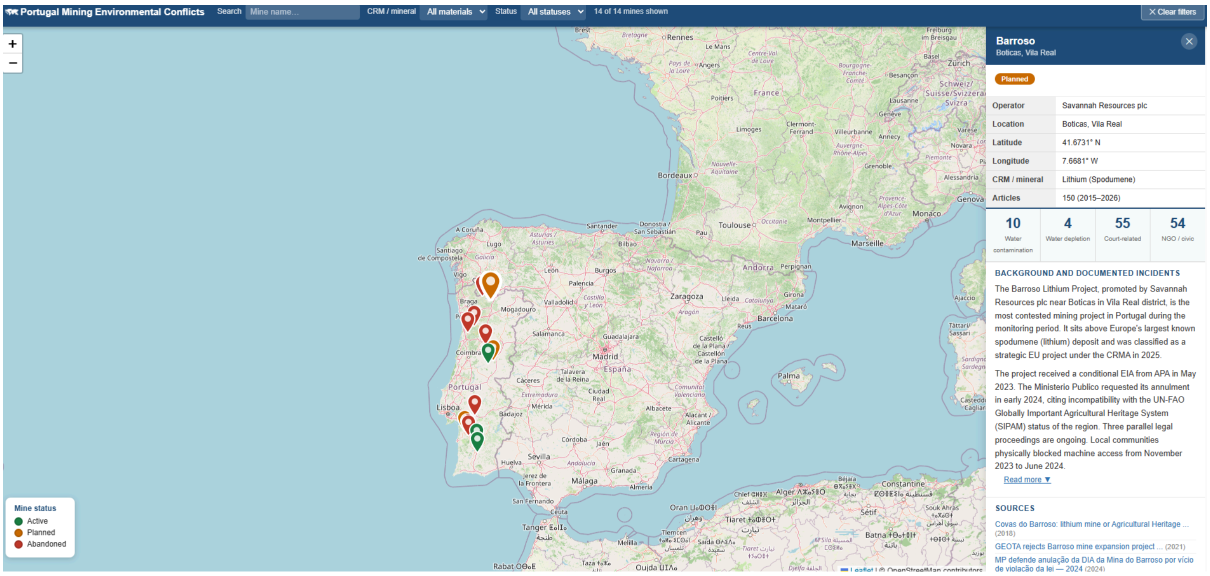


FIGURE 3: Interactive map/database of mines in Portugal. Interactive map available at: https://sergiofspedro.github.io/Miningeurope/

Legacy contamination: the absent operator problem

Seven of the 14 sites in this report are abandoned mines. In each case, the private operators who profited from extraction are no longer present; the costs of remediation fall entirely on the Portuguese state, primarily through EDM and European structural funds. At Azeméis, a 4.6 million euro remediation tender was launched by EDM in October 2025, 67 years after extraction ceased. At Urgeiriça, uranium mine rehabilitation has been ongoing since 2005, with additional complementary remediation phases still underway. At Terramonte, Lousal and Caveira, state-funded remediation has reduced but not eliminated contamination of Douro and Sado basin watercourses. The cumulative cost of remediating Portugal's 199 abandoned mines has exceeded 200 million euros over 25 years of EDM intervention, with ongoing works financed through successive EU structural fund programmes (QCA III, FEDER, Norte 2030, Portugal 2030).

This pattern raises a governance question that is not addressed by current EU frameworks: the absence of mandatory financial guarantees sufficient to cover full remediation costs means that the environmental and financial liabilities of mining are systematically externalised onto the public. The polluter-pays principle exists in law, but in the context of abandoned mining sites, it has rarely been applied in practice.

Part I - Active mines

Neves-Corvo

Castro Verde, Beja | Active

Project overview

Operator	Somincor / Boliden Group
Location	Castro Verde, Beja
Latitude	37.6447 deg N
Longitude	8.1256 deg W
CRM / mineral	Copper|Zinc|Lead|Silver|Tin
Status	**Active**
Articles found	82 (2015-2026)
Years with coverage	1988, 2012, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026

Monitoring summary (2015-2026)

0 Water contamination related articles	**0** Water depletion related articles	**8** Report to police authorities/ Report to police authorities/ Court-related articles	**10** NGO / civic movement related articles

Background and documented incidents

Neves-Corvo is one of the most important copper and zinc mines in Europe, operated by Somincor (a subsidiary of the Boliden Group) near Castro Verde in the Alentejo region. It has been in continuous operation since 1988 and was classified as a strategic project under the EU Critical Raw Materials Act in 2025. Environmental concerns include the management of the tailings dam at Cerro do Lobo, acid drainage into local watercourses, and the risk of contamination to the Mira river basin.

In 2025-2026, a legal action brought by the public prosecutor (Ministerio Publico) led to the suspension of construction of a 49 MW solar plant adjacent to the mine, promoted by Somincor in partnership with EDP and Greenvolt. The prosecutor argued that the project did not comply with the Municipal Master Plan (PDM) of Castro Verde. The case raised wider questions about cumulative environmental impact and water resource management in the Castro Verde area. A subcontractor also filed a claim of 5.6 million euros against Somincor for alleged breach of contract.

The mine's EIA processes (including the fourth raising of the tailings dam at Cerro do Lobo) have been subject to review by the APA (Agencia Portuguesa do Ambiente), with conditions imposed regarding surface and groundwater quality monitoring.

Water-related breach / non-compliance excerpts

The following excerpts from monitored sources reference breaches, infringements or non-compliance with water quality standards:

> *"Environmental concerns at Neves-Corvo include the management of the tailings dam at Cerro do Lobo, acid mine drainage into local watercourses, and the risk of contamination of the Mira river basin. The EIA for the fourth raising of the tailings dam (Barragem de Rejeitados - 4a Fase, cota 255m) was evaluated under DL 197/2005, with conditions imposed on surface and groundwater quality monitoring."*

Source (2016): https://siaia.apambiente.pt/AIADOC/AIA1714/rnt1714201641311419.pdf

> *"A legal action brought by the public prosecutor led to the suspension of construction of a 49 MW solar plant adjacent to the mine on grounds that the project did not comply with the municipal land-use plan (PDM). The case raised questions about cumulative environmental impact and water resource management in the Castro Verde area, particularly regarding the proximity to the Mira river basin."*

Source (2026): https://observador.pt/2026/03/27/central-solar-das-minas-de-neves-corvo-impugnad

Sources

- Lithium projects in Portugal receive classification ... (2025) - https://www.dn.pt/economia/bruxelas-d%C3%A1-for%C3%A7a-m%C3%A1xima-e-classifica-
- Castro Verde: Subcontractor demands 5.6 million more from Somincor ... (2026) - https://lidadornoticias.pt/castro-verde-subempreiteira-exige-a-somincor-mais-56-
- Contested solar plant exempted from environmental assessment (2026) - https://observador.pt/2026/03/27/central-solar-das-minas-de-neves-corvo-impugnad
- Neves-Corvo invests in expansion and longevity of ... (2026) - https://sapo.pt/artigo/neves-corvo-investe-na-expansao-e-longevidade-da-extracao
- The concessionaire company of the Neves-Corvo mine, ... (2026) - https://sapo.pt/artigo/concessionaria-reforca-investimento-na-exploracao-na-mina
- Company demands 5.6 million euros from the owner of the mines of ... (2026) - https://www.jn.pt/pais/artigo/empresa-exige-56-milhoes-de-euros-a-dona-das-minas

Aljustrel

Aljustrel, Beja | Active

Project overview

Operator	Almina - Minas do Alentejo, S.A.
Location	Aljustrel, Beja
Latitude	37.8826 deg N
Longitude	8.1611 deg W
CRM / mineral	Copper|Zinc|Lead|Silver
Status	**Active**
Articles found	54 (2015-2026)
Years with coverage	2014, 2015, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026

Monitoring summary (2015-2026)

0 Water contamination related articles	**1** Water depletion related articles	**5** Report to police authorities/ Report to police authorities/ Court-related articles	**4** NGO / civic movement related articles

Background and documented incidents

Aljustrel, operated by Almina - Minas do Alentejo, S.A., is one of the oldest continuously mined sites in the world, located on the Iberian Pyrite Belt in Beja district. The mine extracts copper, zinc, lead and silver concentrates. It is situated approximately 20 km north of Neves-Corvo, in the same hydrographic basin.

Acid mine drainage (AMD) from the sulphide mineralogy of the ore generates acidic leachate carrying heavy metals (copper, zinc, lead, arsenic) into the local drainage network, with potential long-term impacts on the Sado river basin. Academic studies published by LNEG (Laboratorio Nacional de Energia e Geologia) have documented the geochemical evolution of tailings and drainage in the Aljustrel area.

In 2022, a fatal accident at the mine killed one worker. Subsequent investigations by the Authority for Work Conditions (ACT) identified non-compliance with occupational safety obligations. ALMINA was subject to administrative proceedings. In 2024, approximately 40 workers were temporarily trapped underground following an incident, though all were evacuated safely. Environmental monitoring obligations under the mine's operating licence cover both surface water and groundwater quality in the surrounding area.

Water-related breach / non-compliance excerpts

The following excerpts from monitored sources reference breaches, infringements or non-compliance with water quality standards:

> *"Acid mine drainage (AMD) from the Iberian Pyrite Belt mines, including Aljustrel, has been documented affecting local watercourses and groundwater. The sulphide mineralogy of the ore generates acidic leachate carrying heavy metals (copper, zinc, lead, arsenic) into the drainage network, with potential impacts on the Sado river basin."*

Source (2020): https://www.lneg.pt/wp-content/uploads/2020/03/85_1826_ART_C

> *"Following a fatal accident at Aljustrel mine, investigations by the Authority for Work Conditions (ACT) identified non-compliance with safety and environmental management obligations. ALMINA was subject to administrative proceedings."*

Source (2022):
https://www.facebook.com/cmjornal/videos/um-morto-e-um-ferido-grave-em-acidente-na-mina-de

Sources

- The sad deaths at the Aljustrel mines (2019) - https://visao.pt/atualidade/sociedade/2019-08-10-as-tristes-mortes-das-minas-de-
- Environmental monitoring commission - Aljustrel - Almina (2022) - https://www.almina.pt/pt/sustentabilidade/ambiente-e-natureza
- Download of mineral resources information (2022) - https://geoportal.lneg.pt/pt/dados_abertos/info_recursosminerais/
- Aljustrel mines - Wikipedia, the free encyclopaedia (2022) - https://pt.wikipedia.org/wiki/Minas_de_Aljustrel
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- Alcácer was receiving poisoned water from three mines (2020) - https://osaldahistoria.blogs.sapo.pt/alcacer-recebia-agua-envenenada-por-53284
- People - Almina (2022) - https://www.almina.pt/pt/pessoas
- Accident claims one fatality at Aljustrel mine (2019) - https://www.rtp.pt/noticias/pais/acidente-faz-uma-vitima-mortal-em-mina-de-aljus

Panasqueira

Covilhã / Fundão, Castelo Branco | Active

Project overview

Operator	Beralt Tin & Wolfram (Portugal) S.A. / Almonty Industries Inc.
Location	Covilhã / Fundão, Castelo Branco
Latitude	40.0333 deg N
Longitude	7.75 deg W
CRM / mineral	Tungsten (Wolframite)|Tin|Copper
Status	Active
Articles found	16 (2015-2026)
Years with coverage	2013, 2014, 2016, 2022, 2023, 2024, 2025, 2026

Monitoring summary (2015-2026)

4 Water contamination related articles	**0** Water depletion related articles	**2** Report to police authorities/ Report to police authorities/ Court-related articles	**3** NGO / civic movement related articles

Background and documented incidents

The Panasqueira mine, operated by Beralt Tin and Wolfram (Portugal) S.A. (majority-owned by Almonty Industries Inc.), is one of Europe's most important tungsten producers and has been active since 1896. Located between the Covilha and Fundao municipalities of Castelo Branco district, it extracts tungsten (wolframite), tin and copper.

In January 2026, heavy rains caused the partial collapse of an inactive tailings dam, sweeping toxic waste including lead and arsenic into the Cebola stream, a tributary of the Zezere river. The Quercus and MiningWatch Portugal environmental organisations issued formal alerts to the competent authorities. Water abstraction for local supply was temporarily suspended by the municipality of Covilha, which supplied the affected population by tanker. Municipal analysis did not confirm contamination, but the company acknowledged in 2025 that it was discharging untreated effluent into a tributary of the Zezere.

Scientific studies since 2015 have repeatedly documented elevated arsenic levels in the Zezere river near the mine. A 2019 study by researchers from the University of Porto evaluated the risks of arsenic-contaminated water consumption in the region, noting that the Zezere feeds the Castelo do Bode reservoir, which supplies Lisbon and dozens of municipalities. A doctoral thesis defended at the University of Coimbra in 2015 recorded acid pH and heavy metal levels above recommended limits in Zezere waters near the mining area. In 2025, SIC Noticias reported that discharges from Panasqueira into the Zezere were ongoing and that monitoring data raised public health concerns.

Water-related breach / non-compliance excerpts

The following excerpts from monitored sources reference breaches, infringements or non-compliance with water quality standards:

> *(...) collapse of a slurry dam; toxic waste containing lead and arsenic washed into the Cebola stream, a tributary of the Zêzere; water abstraction suspended; company admits to untreated discharges"*
>
> *Source (2026):*
> https://www.publico.pt/2026/03/24/azul/noticia/minas-panasqueira-quercus-denuncia-risco-co

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- Collapse of tailings dumps at Panasqueira mines ... (2026) - https://sapo.pt/artigo/desabamento-de-escombreiras-nas-minas-da-panasqueira-gera
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- Researchers recover metals from the waste of ... (2025) - https://noticiasdacovilha.pt/investigadores-recuperam-metais-nos-residuos-da-pan
- Panasqueira mines: discharges into the Zêzere river are ... (2025) - https://sicnoticias.pt/pais/ambiente/2025-03-31-video-minas-da-panasqueira-desca
- Panasqueira mine: with over a century of history ... (2022) - https://www.ambientemagazine.com/minas-da-panasqueira-com-mais-de-um-seculo-de-h
- Health risks arising from work in mines (2014) - https://www.mineralex.net/riscos-para-a-saude-decorrentes-do-trabalho-nas-minas-
- Assessment and mitigation of underground environmental risk ... - APRH (2024) - https://aprh.pt/en/other-articles/avaliacap-e-mitigacao-do-risco-ambiental-subte

Part II - Planned

Barroso

Boticas, Vila Real | Planned

Project overview

Operator	Savannah Resources plc
Location	Boticas, Vila Real
Latitude	41.6731 deg N
Longitude	7.6681 deg W
CRM / mineral	Lithium (Spodumene)
Status	Planned
Articles found	162 (2015-2026)
Years with coverage	2012, 2013, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026

Monitoring summary (2015-2026)

10 Water contamination related articles	**4** Water depletion related articles	**55** Report to police authorities/ Report to police authorities/ Court-related articles	**55** NGO / civic movement related articles

Background and documented incidents

The Barroso Lithium Project, promoted by Savannah Resources plc near Boticas in Vila Real district, is the most contested mining project in Portugal during the monitoring period. It sits above Europe's largest known spodumene (lithium) deposit and was classified as a strategic EU project under the Critical Raw Materials Act (CRMA) in 2025.

The project received a conditional Environmental Impact Declaration (EIA) from APA in May 2023 - the first lithium project in Portugal to do so. The EIA conditions included explicit requirements for the protection of watercourses and groundwater in the Barroso-Alvao region. However, the Ministerio Publico (public prosecutor) requested its annulment at the Tribunal Administrativo e Fiscal de Mirandela in early 2024, on grounds of "violation of law," citing incompatibility with the UN-FAO Globally Important Agricultural Heritage System (SIPAM) status of the region and inadequate assessment of water resource impacts.

Three parallel legal proceedings are ongoing: (1) an action by the Junta de Freguesia de Covas do Barroso to cancel the exploration licence; (2) an action by the Comunidade de Baldios de Covas do Barroso against Savannah and private landowners for alleged unlawful occupation of communal land; and (3) a claim by Savannah for access to land for exploratory works. Local communities physically blocked machine access to the site from November 2023 to June 2024. In February 2026, the association Unidos em Defesa de

Covas do Barroso (UDCB) and the environmental law organisation ClientEarth lodged a challenge at the EU Court of Justice (TJUE), arguing the European Commission failed to reassess the project after new evidence of environmental and social risks emerged. The case could set a precedent for other mining projects with strategic EU status.

Five mining concession contracts were signed by the Portuguese government in late 2024, expanding the concession area by 1,139 hectares. The ZERO association criticised this process as lacking transparency and public participation. Key water concerns documented include the impact of open-pit extraction on surface watercourses and groundwater feeding the Tamega and Cavado river basins, and the management of approximately 16 million tonnes of tailings.

Water-related breach / non-compliance excerpts

The following excerpts from monitored sources reference breaches, infringements or non-compliance with water quality standards:

> *"The Public Prosecutor's Office (MP) considers that the Environmental Impact Statement (EIS) for the Barroso lithium mine 'is vitiated by a breach of the law' and 'should be annulled'. The MP warns of the risk of the Barroso-Alvao region losing its World Agricultural Heritage Site (SIPAM) status, which is incompatible with mining operations. The EIS includes explicit conditions for the protection of the region's watercourses and aquifers."*

Source (2024):
https://rr.pt/noticia/pais/2024/02/09/minas-de-litio-boticas-aplaude-anulacao-de-declaraca

> *"The EIA conditions for the Barroso lithium project include explicit requirements for the protection of watercourses and groundwater in the Barroso-Alvao region, whose streams feed the Tamega and Cavado river basins. Environmental groups and the UN Special Rapporteur on Human Rights and the Environment highlighted the risk of water resource depletion from the proposed open-pit extraction."*

Source (2024):
https://novabhre.novalaw.unl.pt/a-exploracao-de-litio-na-mina-do-barroso-minimizacao-de-im

> *"The association Unidos em Defesa de Covas do Barroso (UDCB) and ClientEarth lodged an appeal with the Court of Justice of the European Union (CJEU) in February 2026, challenging the project's classification as strategic under the European Critical Raw Materials Regulation. The appellants argue that the European Commission failed to reassess the environmental and social risks of the project, including the hydrological impacts."*

Source (2026):
https://www.euronews.com/2026/02/09/sacrifice-zones-or-clean-energy-future-eu-court-weighs

Sources

- Borralha mine to have underground exploitation in former ... (2025) - https://www.rtp.pt/noticias/economia/mina-da-borralha-tera-exploracao-subterrane
- Borralha mine to have underground exploitation in the former ... (2025) - https://www.publico.pt/2025/10/07/azul/noticia/mina-borralha-tera-exploracao-sub
- Borralha mine to have underground exploitation in former ... (2025) - https://www.cmjornal.pt/sociedade/detalhe/mina-da-borralha-tera-exploracao-subte
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- Association warns of risks of mining exploitation in ... (2025) - https://www.cmjornal.pt/sociedade/detalhe/associacao-alerta-para-riscos-da-explo

- GEOTA rejects Barroso mine expansion project ... (2021) - https://www.geota.pt/blogs/geota-chumba-projeto-de-ampliacao-da-mina-do-barroso-
- Barroso mine raises sustainability and risk alerts ... (2025) - https://www.supplychainmagazine.pt/2025/06/12/mina-do-barroso-levanta-alertas-de
- Covas do Barroso: lithium mine or Agricultural Heritage ... (2018) - https://www.esquerda.net/artigo/covas-do-barroso-mina-de-litio-ou-patrimonio-agr

Romano de Tresminas

Montalegre / Vila Pouca de Aguiar, Vila Real | Planned

Project overview

Operator	Lusorecursos Portugal Lithium, Lda.
Location	Montalegre / Vila Pouca de Aguiar, Vila Real
Latitude	41.75 deg N
Longitude	7.7167 deg W
CRM / mineral	Lithium (Lepidolite|Spodumene)
Status	**Planned**
Articles found	83 (2015-2026)
Years with coverage	2012, 2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025, 2026

Monitoring summary (2015-2026)

0 Water contamination related articles	**2** Water depletion related articles	**18** Report to police authorities/ Report to police authorities/ Court-related articles	**14** NGO / civic movement related articles

Background and documented incidents

The Romano de Tresminas Lithium Project, promoted by Lusorecursos Portugal Lithium, Lda., is located near Montalegre and Vila Pouca de Aguiar in Vila Real district, targeting lithium from lepidolite and spodumene. It was classified as an EU strategic project in 2025. The site takes its name from the ancient Roman gold mines of Tresminas, a recognised archaeological heritage site.

The project received a conditional EIA in 2023 and five mining concession contracts were signed in late 2024. The EIA public consultation received over 1,000 public submissions - one of the highest participation rates in Portuguese environmental assessment history. The Camara de Montalegre filed legal proceedings to halt the project. A report by journalists linked to the "Operacao Influencer" investigation (which led to the resignation of Prime Minister Antonio Costa) implicated one company as a suspected beneficiary, though Romano's concession process was not the direct subject of criminal charges.

Key water concerns documented include: impacts on mountain streams in Tras-os-Montes that feed the Cavado and Tamega basins; the generation of large volumes of toxic tailings from lepidolite processing; and the absence of a functioning oversight commission (Comissao de Acompanhamento), a legal requirement under the EIA conditions that had not been constituted by mid-2025. The PSD parliamentary group requested the government to accelerate the formation of these oversight commissions.

Lusorecursos has presented the project at international mining conferences (PDAC 2026) as a strategic European lithium source, while civic groups including the movement "Nao as Minas de Montalegre" continue to contest it on social media and in court.

Water-related breach / non-compliance excerpts

The following excerpts from monitored sources reference breaches, infringements or non-compliance with water quality standards:

> *'(...) exploration of Lithium Deposits and Associated Minerals – Romano" to the APA, but the authority clarified that "the EIA for the lithium exploration project in Montalegre had not been processed" due to non-compliance with certain conditions, such as the absence of a document relating to cross-border impacts. On 14 December 2020, the Environmental Impact Assessment (EIA) procedure relating to the Romano mine began (...)"*

Source (2023):
https://ominho.pt/a-polemica-mina-de-litio-em-montalegre-que-levou-a-demissao-de-antonio-c

> *(...) There is no suggestion, either explicit or implicit, that Correio da Manhã had any prior knowledge of such content, let alone that it agrees with, endorses or supports its content. Conduct In the event of a breach of these Rules, Correio da Manhã may suspend the ability to comment for a fixed or indefinite period, or even permanently ban it, regardless of whether the user is a subscriber to Correio da Manhã Premium or to (...)*

Source (2022):
https://www.cmjornal.pt/sociedade/detalhe/mina-de-litio-causa-poluicao-e-ruido-em-montaleg

Sources

- Lusorecursos confident of 'go-ahead' for mine in Montalegre ... (2023) - https://eco.sapo.pt/especiais/lusorecursos-confiante-no-ok-para-a-mina-em-montal
- Lusorecursos confident of 'go-ahead' for mine in Montalegre ... (2023) - https://cnnportugal.iol.pt/lusorecursos/montalegre/lusorecursos-confiante-no-ok-
- Lithium. Company admits "minimal" negative impacts (2020) - https://observador.pt/2020/01/25/empresa-com-a-concessao-admite-impactos-negativ
- Portuguese Environment Agency approves exploitation of ... (2023) - https://sicnoticias.pt/economia/2023-09-07-Agencia-Portuguesa-do-Ambiente-viabil
- Court rejects Montalegre's request to block ... (2024) - https://www.ambienteonline.pt/noticias/tribunal-rejeita-pedido-de-montalegre-par
- Mining history (2024) - https://lusorecursos.com/minaromano.html
- Sao Sebastiao da Pedreira parish (2020) - https://www.patriarcado-lisboa.pt/site/index.php?tem=65&value=open:1698
- The controversial lithium mine in Montalegre that led to ... (2023) - https://ominho.pt/a-polemica-mina-de-litio-em-montalegre-que-levou-a-demissao-de

Lagoa Salgada

Grândola / Alcácer do Sal, Setúbal / Beja | Planned

Project overview

Operator	Redcorp - Empreendimentos Mineiros, Lda. / Ascendant Resources
Location	Grândola / Alcácer do Sal, Setúbal / Beja
Latitude	38.2167 deg N
Longitude	8.5667 deg W
CRM / mineral	Copper|Lead|Zinc|Gold|Silver
Status	**Planned**
Articles found	20 (2015-2026)
Years with coverage	2021, 2022, 2025, 2026

Monitoring summary (2015-2026)

14 Water contamination related articles	**7** Water depletion related articles	**6** Report to police authorities/ Report to police authorities/ Court-related articles	**10** NGO / civic movement related articles

Background and documented incidents

The Lagoa Salgada project, promoted by Redcorp - Empreendimentos Mineiros, Lda. (a subsidiary of Ascendant Resources, Toronto), was a proposed underground polymetallic mine in Grandola and Alcacer do Sal municipalities, targeting copper, lead, zinc, gold and silver. It was classified as a Project of National Interest (PIN) in 2022.

In January 2026, the APA issued a negative EIA decision, rejecting the project specifically on grounds of risks to water resources - the first major Portuguese mining project to be rejected on explicit water grounds. The project would have used sodium cyanide in ore processing and required an average of 864 cubic metres per day of groundwater in one of Portugal's driest regions (Sado/Mira hydrographic network, RH6). The sole water abstraction point supplying local villages (Agua Derramada and Cilha do Pascoal) was located within the proposed mining area.

Environmental groups (ZERO, Quercus, Proteger Grandola), farmers and local authorities had filed extensive objections during public consultation, which closed in November 2025. The municipality of Grandola expressed satisfaction at the APA's decision. The Ascendant Resources board indicated it would seek to appeal or resubmit.

Scientific context for the rejection included references to the legacy of the nearby Lousal and Caveira abandoned mines, which demonstrate the long-term acid drainage and metal contamination risks of pyrite belt mining in the Sado watershed. The Lagoa Salgada case is considered a landmark decision in Portuguese environmental law regarding mining and water resources.

Water-related breach / non-compliance excerpts

The following excerpts from monitored sources reference breaches, infringements or non-compliance with water quality standards:

> *APA rejected the Lagoa Salgada Mine project due to risks to water resources. The project envisaged the consumption of 864 m³/day of groundwater in one of the driest regions of Portugal, within the Sado/Mira river basin (RH6). The sole drinking water catchment point supplying the villages of Água Derramada and Cilha do Pascoal is situated in the centre of the mining area.*

Source (2026):
https://www.ambienteonline.pt/exclusivos/apa-chumba-mina-da-lagoa-salgada-por-riscos-para-

> *"Environmentalists warn that the Lagoa Salgada mine could exacerbate water shortages and cause contamination from heavy metals and sodium cyanide. The project would use sodium cyanide in the processing of the ore, with documented risks of groundwater contamination, similar to that observed in the disused mines of São Domingos and Aljustrel and in the mining areas of the Spanish region of Huelva."*

Source (2025):
https://www.publico.pt/2025/05/23/azul/noticia/mina-lagoa-salgada-processamento-minerio-ou

> *"The Lagoa Salgada mine project, designated as a Project of National Interest (PIN) in 2022, was rejected by the Portuguese Environment Agency in January 2026 specifically on the grounds of risks to water resources. The APA's decision marks the first rejection of a large-scale mining project in Portugal explicitly on the grounds of water protection, and sets a significant legal precedent for future projects in the Iberian Pyrite Belt."*

Source (2025):
https://postal.pt/nacional/mina-da-lagoa-salgada-gera-polemica-por-ameaca-a-recursos-hidri

Sources

- Lagoa Salgada mine project is a danger to resources ... (2025) - https://zero.ong/noticias/projeto-da-mina-da-lagoa-salgada-e-um-perigo-para-os-r
- Lagoa Salgada mine exploitation will use ... (2025) - https://www.publico.pt/2025/05/23/azul/noticia/mina-lagoa-salgada-processamento-
- Lagoa Salgada copper mine drops substance ... (2025) - https://observador.pt/2025/11/14/promotor-da-mina-de-cobre-da-lagoa-salgada-deix
- Zero: Lagoa Salgada mine project is a danger to ... (2025) - https://smart-cities.pt/smn/projeto-da-mina-da-lagoa-salgada-e-um-perigo-para-os
- Lagoa Salgada copper and zinc mine set to be ... (2026) - https://observador.pt/2026/01/23/mina-de-cobre-e-zinco-da-lagoa-salgada-volta-a-
- APA rejects Lagoa Salgada mine project, but ... (2025) - https://www.publico.pt/2025/07/10/azul/noticia/apa-chumba-projecto-mina-lagoa-sa
- Lagoa Salgada copper mine faces environmental setback (2025) - https://observador.pt/2025/07/09/apa-chumba-mina-de-cobre-para-a-lagoa-salgada-p
- Alcácer and Grândola municipalities against copper mine (2025) - https://observador.pt/2025/05/22/autarquias-de-alcacer-e-grandola-contra-mina-de

Argemela

Fundão, Castelo Branco | Planned

Project overview

Operator	Neomina (Almina group)
Location	Fundão, Castelo Branco
Latitude	40.1167 deg N
Longitude	7.5833 deg W
CRM / mineral	Lithium|Tin|Tungsten
Status	**Planned**
Articles found	2 (2015-2026)
Years with coverage	2019, 2021

Monitoring summary (2015-2026)

1 Water contamination related articles	**1** Water depletion related articles	**0** Report to police authorities/ Report to police authorities/ Court-related articles	**0** NGO / civic movement related articles

Background and documented incidents

The Argemela project, promoted by Neomina (a subsidiary of the Almina group, also operator of Aljustrel), is a proposed open-pit lithium, tin and tungsten mine in the Serra da Argemela, Fundao municipality, Castelo Branco district. Local opposition has been led by GPSA (Grupo de Pessoas pela Salvaguarda da Argemela) since 2017.

Key environmental concerns include: the impact of open-pit extraction on the local hydrology of the Serra da Argemela, whose streams feed into the Zezere river system; the risk to water abstractions serving local communities; and the proximity to Castelo do Bode reservoir, which supplies Lisbon. The project was under EIA review during the monitoring period. Opponents have drawn direct comparisons to the Panasqueira mine's documented water quality impacts, noting that the Serra da Argemela is in the same hydrographic sub-basin.

Water-related breach / non-compliance excerpts

The following excerpts from monitored sources reference breaches, infringements or non-compliance with water quality standards:

> *‘impact on the River Zêzere and water abstraction; water consumption for open-cast mining’*
> *Source (2021):* https://siaia.apambiente.pt/AIADOC/DA201/pda201_parecer_ca_2021.pdf

Sources

- Argemela mine - APA - Environmental impact assessment (2021) - https://siaia.apambiente.pt/AIADOC/DA201/pda201_parecer_ca_2021.pdf
- Avelino quarry natural monument (2019) - https://natural.pt/protected-areas/monumento-natural-pedreira-avelino

Part III - Abandoned mines with legacy liabilities

Lousal

Grândola, Setúbal | Abandoned

Project overview

Operator	EDM (remediation) / Museu do Lousal
Location	Grândola, Setúbal
Latitude	38.1 deg N
Longitude	8.4333 deg W
CRM / mineral	Pyrite|Zinc|Lead|Copper (closed 1988)
Status	**Abandoned**
Articles found	10 (2015-2026)
Years with coverage	2017, 2020, 2023, 2024, 2025, 2026

Monitoring summary (2015-2026)

0 Water contamination related articles	**1** Water depletion related articles	**0** Report to police authorities/ Report to police authorities/ Court-related articles	**1** NGO / civic movement related articles

Background and documented incidents

The Lousal mine (closed 1988) in Grandola, Setubal, exploited pyrite, zinc, lead and copper on the Iberian Pyrite Belt. Like Sao Domingos, it left a legacy of acid mine drainage affecting local streams within the Sado basin. EDM has undertaken remediation and the site now also hosts the Lousal Mining Museum (Mina de Ciencia). Historical contamination of local watercourses with heavy metals (arsenic, lead, zinc, copper) has been documented in academic literature. The proximity of the Lousal and Caveira abandoned mines to the proposed Lagoa Salgada project site shaped the debate about cumulative hydrological impacts in the Sado/Mira watershed.

Water-related breach / non-compliance excerpts

The following excerpts from monitored sources reference breaches, infringements or non-compliance with water quality standards:

> *'… "obvious and significant negative effects on the population, the local area, the landscape and the environment – jeopardising the municipality's sustainable development". Following an analysis, it considers that the project "breaches the Municipal Master Plan by disregarding areas protected under the Municipal Ecological Framework, threatens water resources, and poses a risk to public health due to dust, noise and vibrations from explosions".*
> *Source (2025):*
> https://www.publico.pt/2025/05/07/azul/noticia/mina-lagoa-salgada-veneno-mineracao-volta-a

Sources

- The socioeconomic impact of the exploitation of a mineral resource. The case - https://fenix.tecnico.ulisboa.pt/downloadFile/16892449972
- Lagoa Salgada mine: the "poison" of mining returns to ... (2025) - https://www.publico.pt/2025/05/07/azul/noticia/mina-lagoa-salgada-veneno-minerac
- Lousal Living Science Centre - Ciência Viva (2020) - https://lousal.cienciaviva.pt/centro-ciencia-viva-do-lousal/
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- Groups - Lousal Living Science Centre (2017) - https://lousal.cienciaviva.pt/grupos/
- Environmental remediation of the Lousal mining area - EDM (2017) - https://edm.pt/projetos/recuperacao-ambiental-da-antiga-area-mineira-do-lousal/
- Compressor station nucleus, Aljustrel (2024) - https://roteirodasminas.dgeg.gov.pt/pt/lista-de-pontos/nucleo-da-central-de-comp

Caveira

Grândola, Setúbal | Abandoned

Project overview

Operator	EDM (remediation)
Location	Grândola, Setúbal
Latitude	38.1333 deg N
Longitude	8.45 deg W
CRM / mineral	Pyrite|Copper|Zinc (closed 1986)
Status	**Abandoned**
Articles found	4 (2015-2026)
Years with coverage	2024, 2025

Background and documented incidents

The Caveira mine (closed 1986) in Grandola, Setubal, exploited pyrite, copper and zinc on the Iberian Pyrite Belt and left extensive tailings generating ongoing acid mine drainage. EDM is responsible for environmental monitoring and remediation. The site was cited in debates about the cumulative environmental impact of the proposed Lagoa Salgada project on the same watershed. Legacy contamination from Caveira contributes to elevated metal concentrations in streams feeding the Sado basin.

Water-related breach / non-compliance excerpts

The following excerpts from monitored sources reference breaches, infringements or non-compliance with water quality standards:

> *"The Caveira mine (closed 1986) generates ongoing acid mine drainage affecting local streams within the Sado basin. Tailings and waste rock continue to leach heavy metals, with EDM responsible for environmental monitoring. The site was cited in debates about the cumulative hydrological impact of the proposed Lagoa Salgada project on the same watershed."*

Source (2025):
https://www.publico.pt/2025/05/07/azul/noticia/mina-lagoa-salgada-veneno-mineracao-volta-a

Sources

- Bloco de Esquerda requests clarifications from Government ... (2024) - https://www.ambienteonline.pt/destaques/bloco-de-esquerda-pede-esclarecimentos-a

- "It is surreal what is happening at the Caveira mine": levels ... (2024) - https://www.publico.pt/2024/05/19/azul/noticia/irreal-passa-mina-caveira-niveis-
- "'It is surreal what is happening at the Caveira mine': there is ... (2024) - https://www.publico.pt/2024/06/07/azul/direito-de-resposta/irreal-passa-mina-cav
- Caveira mine (2025) - https://www.xn--museu-coleo-martins-da-pedra-lmc3j.pt/mina-da-caveira/

Urgeiriça

Nelas (Canas de Senhorim), Viseu | Abandoned

Project overview

Operator	EDM (rehabilitation) / former ENU
Location	Nelas (Canas de Senhorim), Viseu
Latitude	40.5333 deg N
Longitude	7.85 deg W
CRM / mineral	Uranium (Radium) - abandoned
Status	**Abandoned**
Articles found	2 (2015-2026)
Years with coverage	2021

Monitoring summary (2015-2026)

0 Water contamination related articles	**0** Water depletion related articles	**0** Report to police authorities/ Report to police authorities/ Court-related articles	**1** NGO / civic movement related articles

Background and documented incidents

The Urgeiriça mine in Nelas (Canas de Senhorim), Viseu district, was one of the most important uranium mines in Europe, producing radium (from 1913) and later uranium oxide (U_3O_8, from 1951). It operated under successive state entities until closure. The legacy of radioactive contamination includes elevated uranium and radium concentrations in the Pantanha stream and surrounding groundwater, with potential health implications for local residents.

EDM has carried out extensive remediation since 2005 under a Master Plan, including: slope stabilisation and capping of tailings dumps; construction of a passive mine water treatment system for Shaft 4 (2012) to treat drainage before discharge; sealing of the tailings dam (New Dam); and chemical decontamination of the chemical treatment plant (CTP) and buildings (from 2016 onwards). Former workers and local residents campaigned for recognition as contaminated persons, arguing the state's remediation efforts were insufficient and that the full extent of radiological contamination was not disclosed. Academic analysis documents the political and administrative management of the controversy.

Water-related breach / non-compliance excerpts

The following excerpts from monitored sources reference breaches, infringements or non-compliance with water quality standards:

> *"The Urgeiriça uranium mine left a legacy of radioactive contamination in the Pantanha stream and surrounding groundwater. Remediation works by EDM included construction of a passive mine water treatment system (2012) to treat shaft drainage before discharge, and sealing of the tailings dam. Monitoring has detected elevated uranium and radium in nearby watercourses."*
>
> *Source (2017):* https://edm.pt/en/projetos/environmental-remediation-of-the-urgeirica-mining-area/

Sources

- "We are like lizards": community, struggle and mining ... (2021) - https://www.jornalmapa.pt/2021/03/04/somos-como-os-lagartos-comunidade-luta-e-mi
- "The mine's legacy": two decades of struggle for ... - https://sicnoticias.pt/programas/reportagemespecial/2026-03-14-video-a-heranca-d
- Nuclearity, the work of bodies, and justice: The environmental rehabilitation of the Urgeiriça mines and local protests. (2010) https://journals.openedition.org/spp/277

Borralha

Montalegre, Vila Real | Abandoned

Project overview

Operator	No active operator (inactive since 1986)
Location	Montalegre, Vila Real
Latitude	41.7833 deg N
Longitude	7.95 deg W
CRM / mineral	Tungsten (Wolframite)|Tin
Status	**Abandoned**
Articles found	8 (2015-2026)
Years with coverage	2015, 2024, 2025, 2026

Monitoring summary (2015-2026)

0 Water contamination related articles	**0** Water depletion related articles	**1** Report to police authorities/ Report to police authorities/ Court-related articles	**0** NGO / civic movement related articles

Background and documented incidents

The Borralha mine in Montalegre, Vila Real district, exploited tungsten (wolframite) and tin until its closure in 1986. It is an inactive site with a legacy of tailings and escarpments presenting contamination risks to local watercourses feeding the Cavado river basin. The site re-entered public debate during the 2020s as the Montalegre area became the focus of new lithium mining projects (Romano de Tresminas). A formal public consultation on plans for new exploration at the Borralha site was published on the participa.pt platform in 2025, and an RTP report (October 2025) indicated that underground exploitation of the former mining area was being considered. Local associations and CMJORNAL reported concerns about risks to water quality and biodiversity in the area.

Water-related breach / non-compliance excerpts

The following excerpts from monitored sources reference breaches, infringements or non-compliance with water quality standards:

> *"The abandoned Borralha mine (tungsten and tin, inactive since 1986) in Montalegre has an unresolved environmental legacy including tailings with heavy metal contamination risk to local watercourses. A public consultation on plans for the site was launched in 2025, with local associations raising concerns about water quality impacts on streams feeding the Cávado river basin."*

Source (2025): https://participa.pt/pt/consulta/mina-da-borralha

Sources

- Study confirms severe heavy metal contamination ... (2025) - https://sapo.pt/artigo/estudo-confirma-contaminacao-grave-por-metais-pesados-na-
- APA issues conditional favourable opinion for the ... mine (2026) - https://www.jn.pt/pais/artigo/apa-da-parecer-favoravel-condicionado-a-mina-da-bo
- APA approves Borralha mine project with conditions ... (2026) - https://www.ambienteonline.pt/destaques/apa-aprova-projeto-da-mina-da-borralha-c
- Borralha mines (2024) - https://minob.org/assets/pdf/recursos/MINOB-CaseStudy-MinasDaBorralha-PT.pdf
- Borralha mines - Montalegre (2024) - https://minob.org/portugues/minas-da-borralha.html
- Gold and tungsten in Covas (Vila Nova de Cerveira) - https://www.cm-lousada.pt/cmlousada/uploads/document/file/2515/2674_original.pdf
- The case study of the Borralha mine, Portugal (2015) - https://www.lneg.pt/wp-content/uploads/2020/03/9-Avila-et-al_Metais-pesados-solo
- Impact of deep-sea mining is being assessed (2024) - https://www.sulinformacao.pt/2024/02/impacto-da-mineracao-do-mar-profundo-esta-a

Montemor

Montemor-o-Novo, Évora | Abandoned

Project overview

Operator	Unknown
Location	Montemor-o-Novo, Évora
Latitude	38.65 deg N
Longitude	8.2167 deg W
CRM / mineral	Copper
Status	**Abandoned**
Articles found	6 (2015-2026)
Years with coverage	2023, 2024, 2025

Monitoring summary (2015-2026)

0 Water contamination related articles	**1** Water depletion related articles	**1** Report to police authorities/ Report to police authorities/ Court-related articles	**3** NGO / civic movement related articles

Background and documented incidents

Limited documented information was available for Montemor-o-Novo, Evora. Monitoring identified articles referencing copper extraction activities and associated environmental concerns in the area. At least one court-related article was identified, and NGO or civic involvement was noted. Water impacts from historical copper mining in the Evora region have been documented in academic literature referencing metal contamination of local soils and watercourses. Further manual investigation is recommended to verify the specific incidents identified by the monitoring pipeline.

Water-related breach / non-compliance excerpts

The following excerpts from monitored sources reference breaches, infringements or non-compliance with water quality standards:

> *"Monitoring identified references to copper extraction activities near Montemor-o-Novo with associated environmental concerns. At least one court-related article was identified. Water*

> *impacts from historical copper mining in the Évora region have been documented in academic literature referencing metal contamination of local soils and watercourses."*
> *Source (2016):*
> https://dspace.uevora.pt/rdpc/bitstream/10174/18942/1/livroEVORA_separatas_15-04-16_Parte6

Sources

- Requests (2025) - https://www.dgeg.gov.pt/pt/areas-setoriais/geologia/depositos-minerais-minas/pub
- Mining exploitation does not spare protected areas ... (2025) - https://www.jn.pt/nacional/artigo/exploracao-mineira-nao-poupa-zonas-de-protecao
- Mining's assault on areas with protected status ... (2024) - https://zero.ong/noticias/o-assalto-da-mineracao-as-areas-com-estatuto-de-protec
- Bloco questioned government on gold prospecting in ... (2025) - https://www.esquerda.net/artigo/bloco-questionou-governo-sobre-prospecao-de-ouro
- Contracts in force (2023) - https://www.dgeg.gov.pt/pt/areas-setoriais/geologia/depositos-minerais-minas/pub

Terramonte

Raiva, Castelo de Paiva, Aveiro | Abandoned

Project overview

Operator	MITEL – Minas de Terramonte Lda. (1966-1972, operator); EDM – Empresa de Desenvolvimento Mineiro (remediation, from c.2007)
Location	Raiva, Castelo de Paiva, Aveiro
Latitude	41.0167 deg N
Longitude	8.2500 deg W
CRM / mineral	Lead | Zinc | Silver (Ag-Pb-Zn quartz vein)
Status	**Abandoned**
Articles found	1 (2015-2026)
Years with coverage	2017

Monitoring summary (2015-2026)

1 Water contamination related articles	**0** Water depletion related articles	**0** Report to police authorities/ Report to police authorities/ Court-related articles	**0** NGO / civic movement related articles

Background and documented incidents

The Terramonte mine is located in the parish of Raiva, in the municipality of Castelo de Paiva, Aveiro district . The site exploited a quartz vein with silver, lead and zinc mineralisation (Ag-Pb-Zn). Earliest workings date to the 19th century, reaching depths of approximately 50 metres. The mine was significantly developed in the 1960s by MITEL – Minas de Terramonte Lda., a consortium involving Placer Development Ltd., Compagnie Royale Asturienne des Mines, Noranda Mines Ltd. and Vernon F. Taylor Jr. MITEL presented its first mining plan to the Circunscrição Mineira do Norte in August 1965 and made Terramonte one of the most modern and well-equipped Portuguese mines of its era.

Between 1966 and 1972, Terramonte was one of the most important silver mines in Europe, producing up to 10.3 tonnes of silver in 1967. The mine held estimated total reserves of 427,000 tonnes at 4.16% lead, 3.35% zinc and 224 g/t silver. The ore vein, between 1 and 12 metres wide, contained galena and argentiferous blende, jamesonite, bournonite, pyrargyrite, tetrahedrite, marcasite, argentite, arsenopyrite, pyrite, polybasite, miargyrite and chalcopyrite. Lead concentrate production reached 3,200 t/year (70% Pb and 3,300 g/t Ag) and zinc concentrate 3,200 t/year (50% Zn and 800 g/t Ag). Underground extraction ceased in 1972 and the site was abandoned.

The primary documented environmental impact is the leaching of metals and metalloids — including lead, zinc, silver and arsenic — from tailings into the Ribeira de Terramonte, the Ribeira de Castanheira, and ultimately the Douro river. A 2014 study published in Environmental Earth Sciences (Carvalho et al.) documented metal and metalloid leaching from tailings into streamwater and sediments at the Terramonte site. A professor at the University of Porto's Abel Salazar Biomedical Sciences Institute warned of "the risk of heavy metals in groundwater" at the time of remediation planning. The Quercus association described the situation as potentially involving "very serious situations of environmental contamination and public health risk."

EDM undertook environmental remediation works co-financed by the EU Structural Funds (FEDER / Programa Operacional da Região do Norte, QCA III), with an investment of approximately 1.2 million euros. Works included: construction of peripheral drainage ditches around tailings dumps; collection of clean water from the Ribeira de Terramonte; slope drainage and dissipation channels; waterproofing treatment of modelled tailings surfaces with an impermeable layer, protective fill, topsoil and landscape treatment; and installation of a continuous effluent monitoring and data control system. The remediation reduced contamination levels in the Ribeiras de Terramonte and Castanheira and the Douro river, and improved slope stability and water quality over the medium to long term. Post-remediation monitoring was conducted by the University of Aveiro (Soares, 2014).

Water-related breach / non-compliance excerpts

The following excerpts from monitored sources reference breaches, infringements or non-compliance with water quality standards:

"Metal and metalloid leaching from the Terramonte tailings has been documented affecting streamwater and sediments in the Ribeira de Terramonte and Ribeira de Castanheira, with impacts reaching the Douro river. The EDM remediation works reduced contamination levels and improved slope stability, but the site required continuous effluent monitoring. A risk of heavy metals in groundwater was identified prior to remediation, with the situation described as potentially causing 'very serious situations of environmental contamination and public health risk'."

Sources: Carvalho et al. (2014), Environmental Earth Sciences 71, 2029-2041 | Público (2006): https://www.publico.pt/2006/04/16/jornal/poluicao-provocada-pelas--minas-de-terramonte-resolvida-ate-ao-verao-de-2007-74038 | EDM: https://edm.pt/projetos/recuperacao-ambiental-da-area-mineira-de-terramonte/

Sources

- Terramonte Mine, mindat.org (minerals, geology, references) - https://www.mindat.org/loc-55570.html
- Poluição provocada pelas minas de Terramonte resolvida até ao Verão de 2007 - Público (2006) - https://www.publico.pt/2006/04/16/jornal/poluicao-provocada-pelas--minas-de-terramonte-resolvida-ate-ao-verao-de-2007-74038
- As (nossas) outras minas: de ouro, de Terramonte... - ADEP Castelo de Paiva (2014) - http://adep-paiva.blogspot.com/2014/12/as-nossas-outras-minas-de-ouro-de.html
- Carvalho et al. (2014) - Metal and metalloid leaching from tailings into streamwater and sediments in the old Ag-Pb-Zn Terramonte mine, northern Portugal. Environmental Earth Sciences 71, 2029-2041.

- Environmental remediation of the Terramonte mining area - EDM (2017) - https://edm.pt/projetos/recuperacao-ambiental-da-area-mineira-de-terramonte/

Azeméis

Oliveira de Azeméis, Aveiro | Abandoned

Project overview

Operator	Unknown
Location	Oliveira de Azeméis, Aveiro
Latitude	40.85 deg N
Longitude	8.4667 deg W
CRM / mineral	Arsenic (primary) | Tungsten | Tin | Lead | Silver | Gold | Copper
Status	Abandoned
Articles found	1 (2015-2026)
Years with coverage	2023

Monitoring summary (2015-2026)

1 Water contamination related articles	**0** Water depletion related articles	**0** Report to police authorities/ Report to police authorities/ Court-related articles	**0** NGO / civic movement related articles

Background and documented incidents

The Azeméis mine (officially known as the Minas do Pintor) is located in the parish of Nogueira do Cravo e Pindelo, in the municipality of Oliveira de Azeméis, Aveiro district. The mine was operated by The Anglo-Peninsular Mining & Chemical Co. Ltd from 1875 to 1958, making it one of Portugal's longest-running arsenic extraction operations — and the only mine in Portugal to have commercially extracted arsenic. The site also produced tungsten, tin, lead, silver, gold and copper. The extraction was carried out through a single shaft approximately 204 metres deep, widening into eight galleries. The mine underwent several expansions between 1881 and 1891.

Extraction was halted in 1958 due to the environmental damage caused by the mining operations, and the site was definitively abandoned in 1995 without any remediation or environmental rehabilitation measures being implemented. The abandoned state of the mine left tonnes of arsenic stored in poorly sealed underground galleries. An RTP archive report documented local residents and the public health delegate raising concerns about arsenic contamination immediately after the mine's closure.

The principal documented environmental impact is acid mine drainage from the flooded galleries, which generates contaminated outflow into the Ribeira do Pintor and subsequently the Antuã river. Academic studies have classified the area as a stressed system due to continuous acid drainage carrying heavy metals (arsenic, lead, copper) into local watercourses. Soil contamination studies commissioned by EDM in 2023 documented the presence of heavy metals in soils, tailings dumps and contaminated materials at the former mining area. Local residents historically reported detecting the smell of arsenic near the mine site.

EDM launched a public tender in October 2025 for the environmental remediation of the Minas do Pintor area, with a base price of 4.6 million euros, financed by European funds (Norte 2030 / FEDER). The works include stabilisation and containment of tailings dumps (particularly the slope facing the Ribeira do Pintor), soil decontamination, erosion control, and ecological and landscape recovery. The municipality of Oliveira de Azeméis has separately commissioned the design of a Minas do Pintor Interpretative Centre, with plans to convert the area into a cultural and leisure destination, with an estimated investment of 5 million euros.

Water-related breach / non-compliance excerpts

The following excerpts from monitored sources reference breaches, infringements or non-compliance with water quality standards:

"The closure of extraction of ore in 1958 is associated with the environmental damage caused by this mining operation, which was definitively closed in 1995 without any remediation or environmental rehabilitation measures. The abandoned galleries contain stored tonnes of arsenic. Acid drainage from the flooded galleries reaches the Ribeira do Pintor and the Antuã river, generating continuous discharge of toxic substances into local watercourses."
Source (2023):
https://www.noticiasdeaveiro.pt/estudo-analisa-solos-e-materiais-contaminados-das-antigas-minas-do-pintor/

Sources

- Oliveira de Azeméis: Study analyses soils and materials ... (2023) - https://www.noticiasdeaveiro.pt/estudo-analisa-solos-e-materiais-contaminados-da
- Remediação ambiental das antigas minas do Pintor em concurso público (2025) - https://www.noticiasdeaveiro.pt/remediacao-ambiental-das-antigas-minas-do-pinto-em-concurso-publico/
- Minas do Pintor (mindat.org) - https://www.mindat.org/loc-46297.html
- Há recursos minerais em Oliveira de Azeméis? Jovens Repórteres para o Ambiente (2015) - https://jra.abaae.pt/plataforma/artigo/ha-recursos-minerais-em-oliveira-de-azemeis/
- Minas do Pintor ao abandono - RTP Arquivos - https://arquivos.rtp.pt/conteudos/minas-do-pintor-ao-abandono/
- Quatro antigas áreas mineiras do Norte com recuperação ambiental de 19 milhões - Porto Canal (2025) - https://portocanal.sapo.pt/noticia/363708